

ASSIGNMENT

Cryptography 15IT327E

Name: ANKITA PAL

Reg. No: RA1711003010611

Batch: 1

Faculty: MS. MAHESHWARI

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Assignment on RSA and Diffie Hellman Key Exchange

Q1. Perform encryption and decryption using the RSA algorithm, for the following.

(a) $p=3$; $q=11$, $e=7$; $M=5$

Ans. Key Generation

1. Selecting prime nos. $p \neq q$, i.e., $p=3$, $q=11$.

2. Calculating $n = p \times q = 3 \times 11 = 33$
 $\boxed{n=33}$

3. Calculate $\phi(n) = (p-1)(q-1)$
 $= (3-1)(11-1) = 2 \times 10 = 20$
 $\therefore \boxed{\phi(n)=20}$

4. Select integer e $\{ \gcd(20, 7) = 1 \}$ ✓
 e is given as 7 $1 < 7 < 20$ ✓
 $\therefore$ Both conditions satisfied $\Rightarrow \boxed{e=7}$

5. Calculate d

$d \equiv e^{-1} \pmod{\phi(n)}$
 $\Rightarrow d \times e \% \phi(n) = 1 \Rightarrow d \times 7 \% 20 = 1 \Rightarrow \boxed{d=3}$

6. Public Key $PV = \{e, n\} = \{7, 33\}$
 Private Key $PR = \{d, n\} = \{3, 33\}$

7. Encryption

PT $M < n$ i.e., $5 < 33$ (M is given as 5)

CT $C = M^e \bmod n$
 $= 5^7 \bmod 33$
 $= 78,125 \bmod 33$

$C = 14$

8. Decryption

$C = 14$

PT $M = C^d \bmod n$
 $= 14^3 \bmod 33$
 $= 2744 \bmod 33$

$M = 5$

Hence, M is verified as given.

(b) $p = 5, q = 11, e = 3; M = 9$

Ans: Key Generation

1. selecting p, q , i.e. $p = 5, q = 11$

2. calculating n , i.e., $n = p \times q = 5 \times 11 = 55$
 $n = 55$

3. $\phi(n) = (p-1)(q-1)$
 $\Rightarrow \phi(n) = 4 \times 10 = 40$

4. Selecting e ,
 e given as 3 $\{ \gcd(\phi(n), e) = 1 \Rightarrow \gcd(40, 3) = 1 \checkmark$
 $1 < 3 < 40 \checkmark$
 $\therefore e = 3$

5. ~~Public~~ selecting d and calculating,

$$d = e^{-1} \pmod{\phi(n)}$$

$$\Rightarrow d \times e \pmod{\phi(n)} = 1$$

$$\Rightarrow d \times 3 \pmod{40} = 1$$

$$\Rightarrow \boxed{d = 27}$$

6. Public Key $PV = \{e, n\} = \{3, 55\}$
 $PR = \{d, n\} = \{27, 55\}$

7. Encryption

$$PT \quad M < n, \text{ i.e., } 9 < 55$$

$$CT \quad C = M^e \pmod{n}$$

$$= 9^3 \pmod{55}$$

$$= 729 \pmod{55}$$

$$\Rightarrow \boxed{C = 14}$$

8. Decryption

$$C = 14$$

$$PT \quad M = C^d \pmod{n}$$

$$= 14^{27} \pmod{55}$$

$$\Rightarrow \boxed{M = 9}$$

Hence, M is verified as given.

(6.) $p=7, q=11, e=17; M=8$

Ans: Key Generation

$$1. \quad p=7, q=11$$

$$2. \quad n = 7 \times 11 = 77$$

$$3. \quad \phi(n) = 6 \times 10 = 60$$

$$4. \quad e = 17 \quad \left\{ \begin{array}{l} \gcd(60, 17) = 1 \checkmark \\ 1 < 17 < 60 \checkmark \end{array} \right.$$

$$5. \quad d \times e \pmod{\phi(n)} = 1$$

$$\Rightarrow d \times 17 \pmod{60} = 1$$

$$\Rightarrow 53 \times 17 \pmod{60} = 1$$

$$\Rightarrow \boxed{d = 53}$$

6. Public Key $PV = \{e, n\} = \{17, 77\}$
 Private Key $PR = \{d, n\} = \{53, 77\}$

7. Encryption

PT $M < n$ i.e., $8 < 77$
 CT $c = M^e \bmod n$
 $= 8^{17} \bmod 77$
 $= 57$

8. Decryption

$c = 57$
 PT $M = c^d \bmod n$
 $= 57^{53} \bmod 77$
 $= 8$

Hence, M is verified as given.

(d.) $p = 11$; $q = 13$, $e = 11$; $M = 7$

Ans. Key Generation

- $p = 11$, $q = 13$
- $n = 11 \times 13 = 143$
- $\phi(n) = 10 \times 12 = 120$
- $\boxed{e = 11}$ $\left\{ \begin{array}{l} \gcd(120, 11) = 1 \\ 1 < 11 < 120 \end{array} \right. \checkmark$

5. $d \times e \% \phi(n) = 1$
 $\Rightarrow d \times 11 \% 120 = 1$
 $\Rightarrow \boxed{d = 11}$

6. Public Key $PV = \{e, n\} = \{11, 143\}$
 Private Key $PR = \{d, n\} = \{11, 143\}$

7. Encryption

PT $M < n$ i.e., $7 < 143$
 CT $c = M^e \bmod n$
 $= 7^{11} \bmod 143$
 $= 106$

8. Decryption

$c = 106$
 PT $M = c^d \bmod n$
 $= 106^{11} \bmod 143$
 $= 7$

Hence, M is verified as given.

e. $p=17, q=31, e=7, M=2$

Ans. Key Generation

1. $p=17, q=31$
2. $n=17 \times 31 = 527$
3. $\phi(n) = 16 \times 30 = 480$
4. $e=7 \begin{cases} \gcd(480, 7) = 1 \checkmark \\ 1 < 7 < 480 \checkmark \end{cases}$

5. $d \times e \% \phi(n) = 1$

$\Rightarrow d \times 7 \% 480 = 1$

$\Rightarrow d = 7^{-1} (\text{mod } 480)$

$480 = 7 \times 68 + 4$

$7 = 4 \times 1 + 3$

$4 = 3 \times 1 + 1$

$1 = 4 - 3$

substituting $3 = 7 - 4$

$\Rightarrow 1 = 4 - (7 - 4)$

$\Rightarrow 1 = 4 - 7 + 4$

$\Rightarrow 1 = 2 \times 4 - 7$

sub $4 = 480 - 7 \times 68$

$\Rightarrow 1 = 2(480 - 7 \times 68) - 7$

$\Rightarrow 1 = 960 - 7 \times 136 - 7$

$\Rightarrow 1 = 960 - 7 \times 137$

$\Rightarrow 7^{-1} (\text{mod } 480) \equiv 137$

$\Rightarrow 1 = 2 \times 480 - 136 \times 7 - 7$

$\Rightarrow 1 = 2 \times 480 - 137 \times 7$

$\Rightarrow 7^{-1} (\text{mod } 480) \equiv -137 \Rightarrow 343$

6. Public Key $PU = \{e, n\} = \{7, 527\}$

Private Key $PR = \{d, n\} = \{343, 527\}$

7. Encryption

$$PT \quad M < n, \text{ i.e., } 2 < 527$$

$$\begin{aligned}
 CT \quad c &= M^e \bmod n \\
 &= 2^7 \bmod 527 \\
 &= 128 \bmod 527 \\
 &= 128
 \end{aligned}$$

8. Decryption

$$C = 128$$

$$\begin{aligned}
 PT \quad M &= c^d \bmod n \\
 &= 128^{343} \bmod 527 \\
 &= 2
 \end{aligned}$$

Hence, M is verified as given.

Q2. In a public-key system using RSA, you intercept the ciphertext $C=10$ sent to a user whose public key is $e=5$, $n=35$. What is the plaintext M ?

Ans. Key Generation:

$$C=10 \text{ (Given)}$$

We know that using RSA,

$$\begin{aligned}
 M &= c^d \bmod n \\
 &= 10^d \bmod 35 \quad \text{---(1)}
 \end{aligned}$$

We know that,

$$d \times e \% \phi(n) = 1 \quad \text{---(2)}$$

$$\text{Now, } n=35 \Rightarrow n=7 \times 5 \Rightarrow p=7, q=5$$

$$\therefore \phi(n) = 6 \times 4 = 24 \quad \text{---(3)}$$

Putting $\phi(n)$ & e in (2) we get,

$$d \times 5 \% 24 = 1 \Rightarrow \boxed{d=5}$$

Putting 'd' in (1) we get

$$\therefore M = 10^5 \bmod 35$$

$$\Rightarrow \boxed{M=5} \quad \therefore \text{Plaintext } M \text{ is } \underline{\underline{5}}$$

Q3. Users A and B use the Diffie-Hellman key exchange technique with a common prime $q = 71$ and a primitive root $\alpha = 7$.

- If user A has private key $X_A = 5$, what is A's public key Y_A ?
- If user B has private key $X_B = 12$, what is B's public key Y_B ?
- What is shared secret key?

Ans. 1. select prime number $q = 71$.

$$2. \text{ select } \alpha = 7 \quad \left\{ \begin{array}{l} \alpha < q \\ 7 < 71 \end{array} \right.$$

(a) User - A

$$3. \text{ select } X_A \quad \boxed{X_A = 5} \quad \left\{ \begin{array}{l} X_A < q \\ 5 < 71 \end{array} \right.$$

$$4. \text{ calculate } Y_A = \alpha^{X_A} \bmod q \\ = (7)^5 \bmod 71 \\ = 16807 \bmod 71 = 51$$

$$\boxed{Y_A = 51} \quad \therefore \text{A's public key is } \boxed{51}.$$

(b) User - B

$$5. \text{ select } X_B \quad \boxed{X_B = 12} \quad \left\{ \begin{array}{l} X_B < q \\ 12 < 71 \end{array} \right.$$

$$6. \text{ Calculate } Y_B = \alpha^{X_B} \bmod q \\ = (7)^{12} \bmod 71 = 4$$

$$\boxed{Y_B = 4}$$

$$\therefore \text{B's public key is } \boxed{4}.$$

(c) Key Generation

Sender (A)

$$K = (Y_B)^{X_A} \bmod q \\ = (4)^5 \bmod 71 \\ = \boxed{30}$$

Receiver (B)

$$\begin{aligned}
 K &= (Y_A)^{x_B} \bmod q \\
 &= (51)^{12} \bmod 71 \\
 &= \boxed{30}
 \end{aligned}$$

The shared secret key between sender (A) and Receiver (B) is 30.

Q4. Consider a Diffie-Hellman scheme with a common prime $q = 11$ and a primitive root $\alpha = 2$.

- Show that 2 is a primitive root of 11.
- If user A has public key $Y_A = 9$, what is A's private key x_A ?
- If user B has public key $Y_B = 3$, what is secret key K shared with A?

Ans. Key Generation

a) $(11) = 10$

$$2^{10} = 2^{10} = 1024 \equiv 1 \bmod 11$$

If we check 2^n for $n < 10$, you will find that none of the values is $1 \bmod 11$.

b) 6 because $2 \cdot 6 \bmod 11 = \boxed{9}$

c) $K = 36 \bmod 11 = \boxed{3}$